Date: / /

Experiment No. 15

Aim: To study the pH change in the titration of strong base (0.1M NaOH) and strong acid (0.1M HCl) using universal indicator.

Apparatus: Burette, pipette, conical flask, beakers, burette stand, funnel, beakers, etc.

Chemicals: 0.1M HCl, 0.1M NaOH, universal indicator.

Theory: The solution of strong base NaOH, due to higher concentration of OH^- ions has higher pH. As HCl is added to the NaOH solution, the pH of the solution decreases which changes the color of universal indicator which has a wide range of pH from 0 to 14

Procedure:

1. Wash the burette, pipette and conical flask with water.
2. Rinse the burette with 0.1M HCl solution and fill it with HCl solution after removing air bubble from nozzle and adjust lower meniscus of solution at zero mark.
3. Rinse the pipette with 0.1M NaOH solution and pipette out 10 mL of NaOH and transfer in the conical flask.
4. Add 2 to 3 drops of universal indicator into the solution in conical flask, shake well, observe the violet color of solution and record the pH of the solution.
5. Add a drop of 0.1M HCl solution from the burette in conical flask till the color of the solution changes to indigo color. Note the burette reading.
6. Continue addition of 0.1M HCl solution from the burette dropwise, till the color of the solution changes to blue, then green, yellow, orange and finally red which indicates changes in pH of solution.
7. Note down the burette readings corresponding to each color change.
8. Don't fill the burette with HCl solution till color changes from indigo to red.

Observation:

1. Solution in burette : 0.1 M Hcl
2. Solution in conical flask : 10 mL of 0.1 M NaoH
3. Indicator : universal indicatoe

Observation Table:

Sr. No.	Color of solution	Volume of 0.1M HCl from burette in mL	pH	Nature
1	Indigo	8.7	10.5	Basic
2	Blue	8.65	9.10	Basic
3	Green	8.6	7	Neutral
4	Yellow	8.55	6	Acidic
5	Orange	8.5	4	Acidic
6	Red	0	2	Acidic

Conclusion:

The pH of the solution goes on decreasing on addition of 0.1M HCl from the burette. The different concentration of H^+ ions gives different colors with universal indicator.

Remark and sign of teacher: ....

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. In the titration of strong base 0.1M NaOH aginst 0.1M HCl solution, using universal indicator, the color of the solution changes to blue then pH of the solution is
   a. zero　　b. 4　　c. 7　　d. 8
2. In the titration of strong base (0.1M NaOH) aginst 0.1M HCl solution, using universal indicator, the pH of the solution changes to 6 then the color of the solution in the flask becomes ....
   a. blue　　b. yellow　　c. orange base d. green
3. In the titration of strong base 0.1M NaOH aginst 0.1M HCl solution, using universal indicator the pH of the solution (in the conical flask) becomes 1, the nature of flask solution is ....
   a. strongly acidic　　b. strongly basic　　c. weekly acidic　　d. neutral
4. In the titration of strong base 0.1M NaOH aginst 0.1M HCl solution, using universal indicators, the color of the flask solution becomes green, what is the burette reading of HCl solution?
   a. 10 mL　　b. 8 mL　　c. 12 mL　　d. 6 mL
5. In the titration of strong base 0.1M NaOH aginst 0.1M HCl solution, using universal indicator, the color of the solution in the flask becomes blue then nature of flask solution is
   a. strongly acidic　　b. strongly basic　　c. weakly basic　　d. neutral
6. What is the color of the solution in conical flask, when 10 mL of 0.1M NaOH is neutralized by 0.1M HCl solution?
   a. Blue　　b. Green　　c. Yellow　　d. Orange

Short answer questions

1. Why is in given experiment color of the solution changes?

Ans. The colour changes because the pH of the solution changes during neutralization, and the universal indicator shows different colours at different pH values.

2. What is color of universal indicator in strong acidic and strong basic medium?

Ans. The universal indicator is red in a strong acidic medium and violet/purple in a strong basic medium.

3. What is change in pH of the 0.1M NaOH while adding 0.1M HCl from the burette?

Ans. The pH gradually decreases from a high basic value towards 7 and then becomes acidic as more HCl is added.

4. Why universal indicator is better than litmus paper?

Ans. Universal indicator gives a range of colours corresponding to different pH values, whereas litmus paper only indicates whether a solution is acidic or basic.

5. If the color of universal indicator changes from green to yellow, predict the change in pH.

Ans. The pH decreases from about 7 (neutral) to about 6 (slightly acidic).

Remark and sign of teacher: ....